

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the development, implementation, testing, and evaluation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**. The discussion follows the structure of the Chapter 4 guide while replacing the guide's sample content with the system, processes, technologies, user roles, and database structure documented in Chapters 1-3. The developed system responds to the operational situation of Luna's Arrozcaldo, an eight-branch food business in Iloilo City that previously relied on paper-based sales records, manual inventory monitoring, and physical branch visits for consolidated management information.

The chapter explains how information moves from branch-level sales processing into a centralized PostgreSQL database and then into management dashboards and reports. It also discusses the major modules, database relationships, security controls, deployment arrangement, and testing procedures. Screenshot and diagram locations are provided as placeholders because actual screenshots must come from the final system and should not be fabricated.

The evaluation discussion follows the four criteria established in Chapter 3: functionality, usability, security, and efficiency. The formal respondents were the business owner or administrator, branch managers, and selected cashiers or counter staff. The completed Chapters 1-3 document the evaluation instrument and its administration, but they do not provide the

final item-level weighted means; therefore, this document does not invent numerical ratings.

4.2 Overview of the Developed System

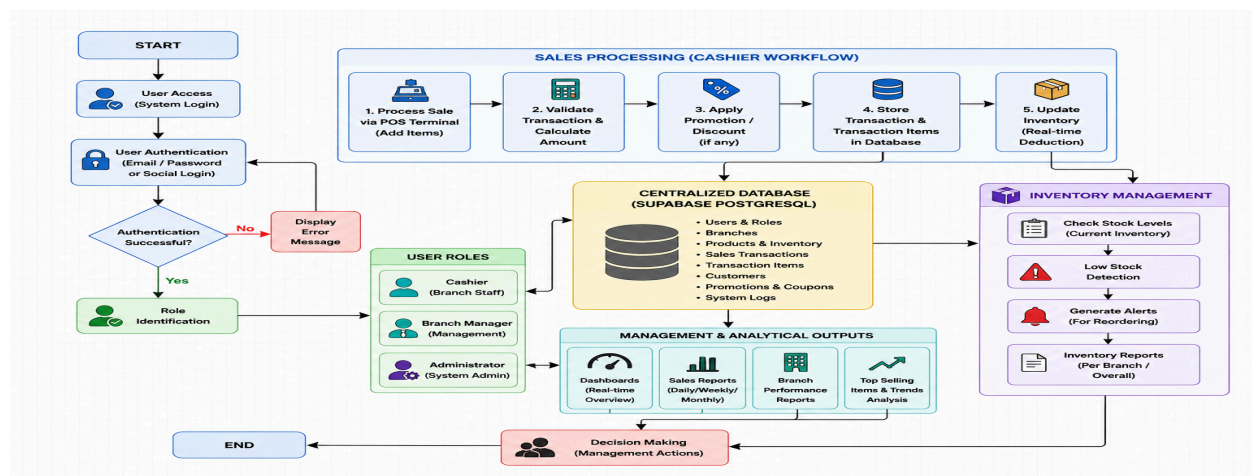
The developed system is a web-based **Digital Sales Processing System** designed specifically for **Luna's Arrozcaldo**. Its primary purpose is to replace paper-based sales recording with a digital point-of-sale process while providing centralized management information for the business's eight branches. The system connects sales processing, inventory, customer information, promotions, and analytical reporting through a centralized PostgreSQL database hosted on Supabase.

At the branch level, the cashier uses the POS terminal to select products, enter quantities, record customer information when applicable, apply valid discounts or promotions, calculate the transaction total, and complete the sale. A completed transaction is recorded in the central database and is associated with the branch and cashier responsible for the transaction. Transaction Items preserve the individual products included in the sale. The completed transaction also causes the corresponding inventory information to be updated.

At the management level, the dashboard and reporting functions organize accumulated transaction data into information that can be reviewed without manually combining paper records. The documented reporting functions include daily, weekly, and monthly sales reports, branch performance comparisons, menu item rankings, trend analyses, inventory status, customer purchase information, and promotional information. These outputs support management decisions using a common data source.

The major objectives addressed by the system are to provide a secure web-based management dashboard, replace paper-based sales recording with a digital POS terminal, centralize information from all eight branches, generate analytical reports and visual summaries, maintain real-time inventory information, support customer profiles and purchase histories, and manage coupons and promotions. These objectives directly correspond to the problems and specific objectives stated in Chapter 1.

Figure 4.1. Overall System Workflow Diagram



The workflow begins with user authentication and role identification. After successful authentication, the authorized user is directed to functions permitted for the assigned role. A cashier processes a sale through the POS terminal; the system validates the transaction, calculates the amount, applies an applicable promotion, stores the transaction and transaction items, and updates inventory. Management users retrieve centralized information through dashboards and reports. The connected records allow one sales event to contribute to operational records and analytical outputs.

4.3 System Development Methodology

The system was developed using the **Agile Development Model** identified in Chapter 3. Agile was selected because the project required repeated interaction with the client, progressive development of system functions, and adjustment of details based on review and feedback. Development was divided into manageable increments, with each sprint producing a working portion of the system that could be reviewed before the next increment.

The development process covered Requirements Analysis, System Design, Development, Testing, Deployment, and Evaluation. Requirements Analysis established the operational problems and system requirements through interviews, observation, and document review. System Design translated those requirements into architecture, workflow, user interface, database, and access-control structures. Development implemented the designed components, while testing examined individual and integrated behavior. Deployment prepared the application for its intended environment, followed by user acceptance and evaluation.

The development process consisted of the following phases:

Requirements Analysis

The requirements analysis phase focused on the actual sales, inventory, customer, and management practices of Luna's Arrozcaldo. The researchers identified difficulties associated with paper-based sales recording, physical branch monitoring, manual inventory checking, limited customer information, and the absence of centralized analytical reporting. These findings became requirements for digital sales processing, centralized storage, inventory management, customer management, promotions, reporting, authentication, and role-based access.

System Design

The system design phase established the architecture, data flow, entity relationships, database normalization, user roles, and interface organization. The multi-branch setting was considered by associating users, transactions, and inventory records with branches. This branch-aware structure permits centralized reporting while preserving the source location of operational records.

Development

Development transformed the approved design into a working web application. PHP was used for server-side application logic, while HTML, CSS, JavaScript, and Bootstrap were used for the web interface. PostgreSQL hosted through Supabase provides centralized data storage. Docker supports a consistent development and runtime environment, Railway is identified for production deployment, and Git supports collaborative version control.

Testing

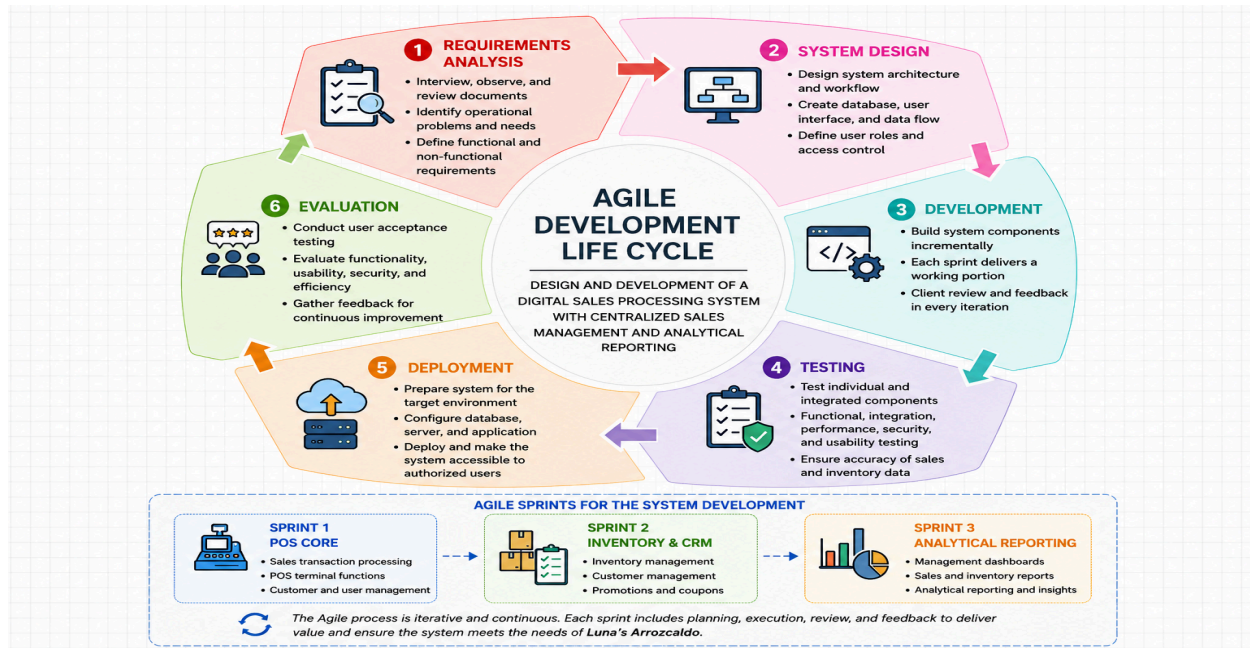
Testing verifies that the system functions correctly and that the different components communicate as intended. The documented testing approach includes functional, integration, performance, security, and usability testing. Particular attention is required for transaction accuracy because an error in sales computation or inventory updating can affect both operational records and management reports.

Deployment and Evaluation

After development and testing, the system was prepared for deployment and user acceptance testing. The intended users were the business owner or administrator, branch managers, and selected cashiers or counter staff. The evaluation instrument

contains twenty Likert-scale items covering functionality, usability, security, and efficiency. Chapter 3 reports twenty-five respondents and complete responses for the twenty core items, while the pilot reliability coefficients ranged from 0.78 to 0.89.

Figure 4.2. Agile Development Life Cycle



The diagram should show the iterative relationship among requirements analysis, system design, development, testing, deployment, and evaluation. Chapter 3 identifies three development sprints: Sprint 1 for POS Core, Sprint 2 for Inventory and CRM, and Sprint 3 for Analytical Reporting.

Sprint-by-Sprint Development Discussion

Sprint 1 focused on the POS core because transaction processing is the central operational function. The increment established the basic authentication and sales workflow needed for a cashier to select products, enter quantities, calculate totals, and store completed transactions. The related user, branch, product,

transaction, and transaction-item relationships were established as part of this core.

Sprint 2 focused on Inventory and Customer Relationship Management. Inventory was connected to completed sales so that stock information could be updated when transactions were processed. The design also included low-stock thresholds and records for deliveries, spoilage, adjustments, and inter-branch transfers. Customer profiles and loyalty information were connected to transaction histories to support repeat-purchase recognition.

Sprint 3 focused on Analytical Reporting. Accumulated transaction data was organized into management reports and visual summaries. The documented requirements include daily, weekly, and monthly sales reports, branch comparisons, menu item rankings, and trend analyses. The dashboard was designed around management questions concerning sales performance, branch differences, inventory alerts, and active promotions.

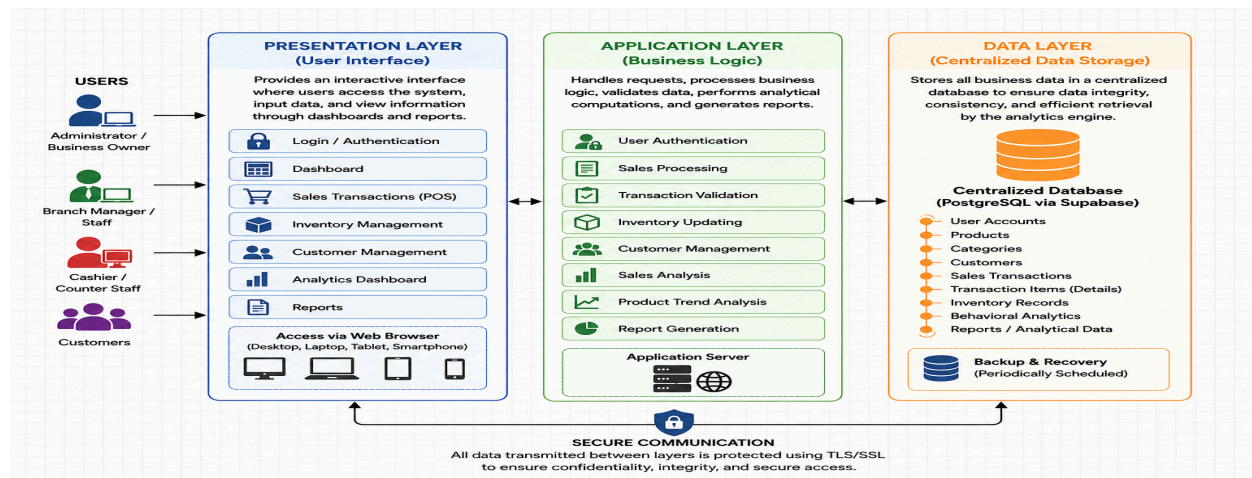
The sprint arrangement also supported risk management. Chapter 3 identifies scheduling delays, scope creep, data loss, low respondent turnout, and connectivity problems as risks. The documented mitigation measures included early scheduling, sprint prioritization, version-controlled commits, automated Supabase backups, respondent planning, and an offline-first POS approach with synchronization on reconnection.

4.4 System Architecture

The system uses a web-based client-server arrangement organized around the user interface, application processing, and centralized data services. This separates presentation from business logic and database operations. Such separation is

appropriate for a multi-branch system because branch users access the application through a browser while the centralized database remains the common repository for sales, inventory, customer, staff, and promotional information.

Figure 4.3. System Architecture Diagram



The architecture should show users accessing the web interface through a modern browser, PHP handling application logic, PostgreSQL hosted on Supabase providing centralized storage, and Railway providing the documented production deployment environment.

The architecture consists of the following components:

4.4.1 Presentation Layer

The presentation layer is the part of the system directly used by personnel. It presents login controls, dashboards, POS functions, inventory information, customer information, promotion functions, and analytical reports according to the user's role. HTML, CSS, JavaScript, and Bootstrap support the interface. Bootstrap 5 provides consistent interface components and responsive presentation.

4.4.2 Application Layer

The application layer contains the business logic that processes user requests. PHP handles authentication and session management, validates submitted information, processes sales, calculates totals, communicates with the database, updates inventory, and prepares information for reporting. Role-based rules are applied so that users receive access appropriate to their responsibilities.

4.4.3 Data Layer

The data layer is provided by PostgreSQL hosted through Supabase. It stores branches, users, products, categories, transactions, transaction items, inventory, customers, loyalty points, promotions, and audit logs. Foreign-key constraints and indexes support data integrity and reporting performance.

4.4.4 Security and Data Integrity Layer

Security is applied across the architecture. Data transmitted through the system is protected using TLS. Passwords are stored as salted cryptographic hashes using PHP's bcrypt implementation. PostgreSQL transaction atomicity protects the consistency of POS operations. Audit logs associate significant events with user and timestamp information, providing accountability.

4.5 Hardware, Software, and Network Requirements

The system is web-based, so branch workstations need a device capable of running a modern browser and maintaining a reliable connection to the deployed application. The POS is intended to run on a desktop or laptop at each branch counter, while

authorized management users can access the web-based dashboard from an appropriate internet-connected device.

Table 4.1. Hardware Requirements

Component	Practical Requirement	Purpose
Computer	Desktop or laptop capable of running a modern browser	Branch POS and management access
Memory	Sufficient RAM for browser and web application use	Normal system operation
Display	Monitor suitable for reading POS and dashboards	Transaction and management viewing
Network equipment	Reliable branch network and internet connection	Connection to centralized services
Peripherals	Standard keyboard and pointing device; compatible output hardware when used	Supports transaction processing

Table 4.2. Software Requirements

Technology	Purpose
PHP	Backend application logic, authentication, transaction processing, database queries, and reports
PostgreSQL	Relational database management

Supabase	Cloud hosting for PostgreSQL and authentication services
HTML/CSS/JavaScript	Web interface structure, styling, and client-side behavior
Bootstrap 5	Responsive and consistent interface components
Docker	Consistent development and runtime environment
Railway	Production application deployment environment identified in Chapter 3
Git	Version control and collaborative development
Modern web browser	Access to the web-based system

4.5.1 Network Requirements

A reliable internet connection is important because the system uses centralized cloud services. The network permits branch POS terminals to communicate with the application and database and allows authorized management users to access consolidated information remotely. Chapter 3 identifies connectivity as a risk and documents an offline-first POS with synchronization on reconnection as a mitigation approach.

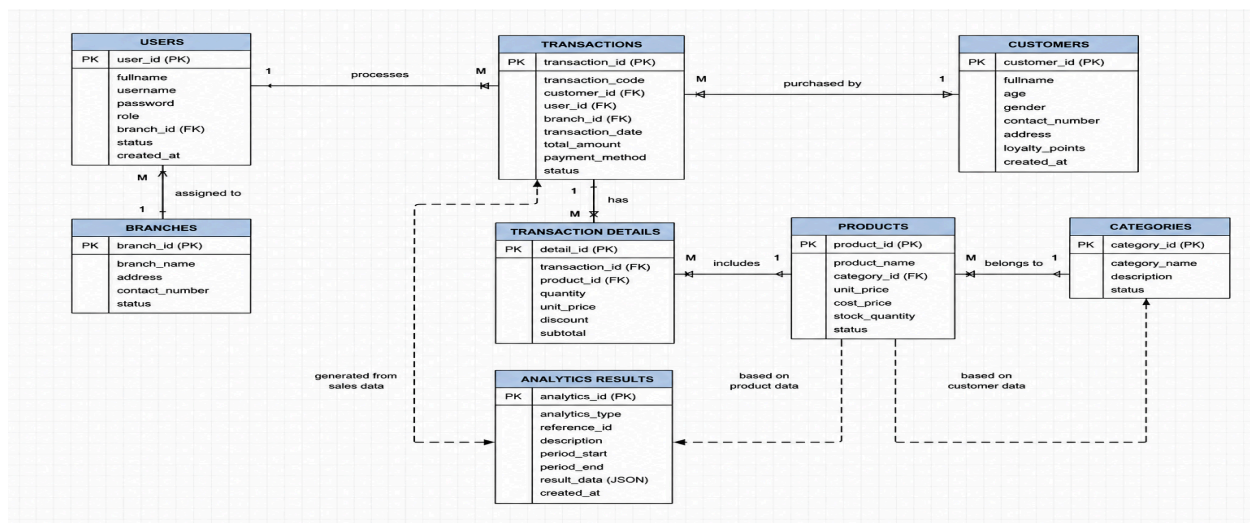
The system does not require a separate database at every branch. Branches access the common web application and centralized database. This reduces duplicate data stores and makes the centralized database the authoritative source of operational information. Secure communication is required whenever

information is transmitted between users, the application, and cloud services.

4.6 Database Design

The database serves as the central storage foundation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting** by organizing sales transactions, customer information, product records, and analytical data. The system uses a **relational database structure** to reduce data redundancy, maintain data integrity, and support efficient retrieval of information required for centralized sales management and analytical reporting. The database design was derived from the system requirements and the relationships among users, customers, products, transactions, transaction details, and analytics.

Figure 4.4. Entity Relationship Diagram



The Entity Relationship Diagram (ERD) presents the relationships among the major entities that support the system's sales processing, customer management, product management, transaction recording, and analytical reporting functions. The primary

entities of the database include **Users, Customers, Products, Transactions, Transaction Details, and Analytics**, which work together to maintain organized operational and analytical records.

The primary entities of the database include:

- **Users** - stores the accounts and role information of authorized system users.
- **Customers** - maintains customer information used to support customer management and behavioral analysis.
- **Products** - stores menu products and their corresponding product information.
- **Transactions** - records completed sales processed through the digital sales processing system.
- **Transaction Details** - records the individual products included in each completed transaction.
- **Analytics** - stores or supports processed analytical information derived from historical transaction data for reporting and decision-making.

Users Table

The Users Table stores the account information of authorized personnel who access and operate the system. It supports user authentication and role-based access by identifying the user and the level of access appropriate to their responsibilities.

Field	Description
user_id	Primary Key
full_name	Complete name of the user
username	Login username

password	Stored password information
role	User access level or assigned responsibility

Customers Table

The Customers Table stores customer information used by the system to maintain customer records and support the analysis of purchasing behavior. These records provide information that can be associated with sales transactions and used as a basis for customer-related analytical reporting.

Field	Description
customer_id	Unique identifier of the customer
fullname	Customer's complete name
age	Customer age
gender	Customer gender

Products Table

The Products Table contains the products or menu items managed by the system. It provides the product information needed during sales processing and allows transaction records to identify which products were purchased.

Field	Description
product_id	Unique identifier of the product
category_id	Identifier of the product category
product_name	Name of the product or menu item
selling_price	Selling price of the product
active_status	Indicates whether the product is currently active

date_added	Date the product was added to the system
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Transactions Table

The Transactions Table records completed sales processed through the system. It serves as the main record of each sales transaction and connects the transaction with the customer and the user responsible for processing it when applicable.

Field	Description
transaction_id	Unique identifier of the transaction
user_id	Identifies the user who processed the transaction
customer_id	Identifies the associated customer, when applicable
transaction_date	Date and time the transaction was recorded
total_amount	Total amount of the completed transaction

Transaction Details Table

The Transaction Details Table records the individual products included in each sales transaction. It provides the detailed connection between transactions and products, allowing the system to determine the items purchased, their quantities, and their corresponding amounts.

Field	Description
transaction_detail_id	Unique identifier of the transaction detail
transaction_id	Identifies the related transaction
product_id	Identifies the purchased product

quantity	Number of units purchased
price	Price of the product at the time of the transaction
subtotal	Calculated amount for the product based on quantity and

Analytics Table

The Analytics Table supports the system's analytical reporting by organizing information derived from historical sales and customer transaction records. The stored analytical information can be used to identify purchasing patterns, frequently purchased products, and sales trends that support management in making data-driven decisions.

Field	Description
analytics_id	Unique identifier of the analytical record
transaction_id	Identifies the transaction used as an analytical reference
product_id	Identifies the product included in the analysis
customer_id	Identifies the customer associated with the analytical record, when applicable
analysis_type	Identifies the type or category of analysis
analysis_result	Stores the resulting analytical information

The relationships among the tables enable the system to track customer purchases, recognize repeated buying patterns, determine product demand, and produce comprehensive analytical reports. Proper indexing and foreign key constraints maintain

data integrity and improve query performance, allowing the system to efficiently handle both sales transactions and analytical processes.

4.7 Module Descriptions

The system is organized into functional modules that correspond to the business processes identified in Chapter 1 and the use cases described in Chapter 3. The modules share the centralized database, allowing a completed sale to affect transaction records, inventory, customer history, loyalty information, promotion records, and analytical outputs. Role-based access controls which functions each user can perform.

4.7.1 Administrator Module

The Administrator module is intended for the business owner or administrator. It supports system-wide oversight and management of users, branches, products, promotions, and centralized business information. Administrative actions are subject to role-based access rules and significant changes can be associated with the responsible account through audit information.

4.7.2 Branch Manager Module

The Branch Manager module supports day-to-day branch oversight. A branch manager can review branch-related sales and inventory information and use reports to understand the performance of the assigned location. The branch-aware database preserves the connection between operational records and their source branch.

4.7.3 Cashier Module

The Cashier module centers on the POS terminal. The cashier selects products, enters quantities, records customer

information when applicable, applies valid discounts or promotions, confirms the total, and completes the transaction. The completed transaction is stored centrally and inventory is updated. This replaces the paper-based sales process.

4.7.4 Inventory Module

The Inventory module provides branch-level stock visibility. It keeps inventory information current as sales are processed and maintains records of deliveries, spoilage, adjustments, and inter-branch transfers. It also provides low-stock alerts when configurable thresholds are crossed.

4.7.5 Sales Processing Module

Sales Processing is the core operational function. The cashier selects products and quantities, reviews the order, and completes the transaction. The application retrieves prices, calculates totals, applies an applicable promotion, and records the sale. The transaction is associated with its branch and cashier and may be associated with a customer.

4.7.6 Customer Management Module

Customer Management supports digital customer profiles and purchase histories. The information provides a basis for identifying repeat customers and managing loyalty points. Personal information is handled in accordance with the Data Privacy Act of 2012 as stated in the study.

4.7.7 Promotions and Coupon Management Module

The Promotions module enables management to create digital discount offers, define their discount type and value, set validity periods, and determine branch scope. A valid promotion can be associated with a transaction during sales processing,

creating a record that can be considered in promotional reporting.

4.7.8 Reporting and Analytics Module

The Reporting and Analytics module transforms stored transaction data into management information. Documented outputs include daily, weekly, and monthly sales reports, branch comparisons, menu item rankings, trend analyses, inventory status, customer purchase behavior, and revenue trends. Visual dashboards make these outputs easier to interpret.

4.7.9 Role-Based Access Control

Role-based access limits functions according to responsibility. The three principal roles are administrator, branch manager, and cashier. The administrator has the broadest management authority, branch managers work primarily with branch operations and management information, and cashiers focus on transaction processing. Authentication verifies credentials before protected functions become available.

Maintenance includes monitoring the system, correcting identified issues, reviewing logs, maintaining backups, and applying necessary updates. Agile also supports maintenance because user feedback can become a future iteration. Changes should be prioritized so that improvements do not compromise sales records, inventory data, or reporting.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion of the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. It brings together the problems identified in Chapter 1, the methodology and technical decisions in Chapter 3, and the implementation and testing discussion in Chapter 4.

The conclusion is based on documented system capabilities. It does not introduce new modules or unsupported claims. Because the supplied Chapters 1-3 do not provide final numerical weighted means, the conclusion does not assign unsupported numerical ratings.

5.2 Summary of Findings

The development and evaluation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo** produced findings aligned with the study objectives:

- The system provided centralized sales management by connecting transactions from Luna's Arrozcaldo's eight branches to a shared database, allowing authorized management users to access consolidated information.
- The digital POS improved sales processing by organizing product selection, quantity entry, total calculation, promotion application, and transaction recording, reducing manual recording steps.

- The centralized sales data supported analytical reporting through daily, weekly, and monthly reports, branch comparisons, product rankings, and sales trend analysis for better decision-making.
- The system integrated inventory management with sales transactions, allowing completed sales to update stock records and supporting low-stock monitoring and other inventory movements.
- Customer and promotion features supported customer management by connecting customer profiles with purchase histories and loyalty points while allowing promotions to include discounts, validity dates, and branch coverage.
- Security and accountability were strengthened through authentication, role-based access, encrypted passwords, TLS, audit logs, database constraints, and automated backups.
- The Agile development approach supported continuous improvement through incremental development of the POS, inventory and customer management, and analytical reporting features, with review and adjustments during development.
- The system was evaluated based on functionality, usability, security, and efficiency, with respondent administration and evaluation procedures documented as part of the system assessment.

5.3 Conclusion

The **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting** provides a practical digital response to the operational problems identified at Luna's Arrozcaldo. The system moves the organization from fragmented

paper-based recording toward a centralized information environment in which transactions, inventory, customer information, promotions, and management reports are connected.

The system addresses the need for improved visibility by allowing authorized management users to access information through a web-based dashboard rather than relying solely on physical collection of branch records. Its branch-aware database preserves the origin of transactions and inventory while allowing information to be consolidated.

The digital POS addresses weaknesses in manual sales recording through a consistent transaction workflow. The system calculates totals, supports promotions, stores transaction details, and connects completed sales with inventory and reporting. This creates more structured and traceable transaction records.

The reporting and analytics component transforms transaction records into organized management information. Daily, weekly, and monthly reports, branch comparisons, menu rankings, and trend analyses allow management to examine business activity from different perspectives.

Inventory, customer, and promotion functions extend the system beyond basic transaction recording. Inventory monitoring supports current stock information and low-stock awareness. Customer profiles and loyalty information provide a basis for relationship management. Promotions provide a structured method for defining discount campaigns.

Security is a necessary part of the conclusion because the system centralizes business records. Authentication, role-based access, bcrypt password hashing, TLS, audit logs, relational integrity constraints, and database backups provide

complementary controls for confidentiality, integrity, accountability, and recovery.

Overall, the project demonstrates how digital technology can be tailored to a heritage food business without changing its established service identity. The system applies technology primarily to internal sales, inventory, customer, promotional, and management processes, strengthening the information infrastructure behind Luna's Arrozcaldo.

5.4 Recommendations

To further improve the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**, the following recommendations are proposed based on the system's current functions, findings, and identified future needs:

System Improvements

- Improve inventory monitoring by adding clearer low-stock alerts to help staff identify products that need replenishment on time.
- Enhance analytical reporting by providing more detailed sales summaries, branch comparisons, product rankings, and sales trends to support management decisions.
- Expand report export options so authorized users can easily save and share reports for business monitoring and record-keeping.
- Strengthen data security and backup procedures to protect sales, customer, inventory, and management records from unauthorized access or data loss.

Future Enhancements

- Develop sales forecasting features that can use accumulated historical sales data to help predict future sales and inventory needs.
- Explore a mobile-friendly management interface to allow authorized managers to conveniently monitor sales and reports using mobile devices.
- Consider additional analytical features that can provide deeper insights into customer purchasing behavior, product performance, and branch sales trends.

Implementation Strategies

- Provide proper training for staff and managers so they can correctly use the POS, inventory, customer management, and reporting functions.
- Conduct regular system maintenance and data backups to keep the system reliable, secure, and updated.
- Continue gathering user feedback from cashiers, branch managers, and administrators to identify areas that can be improved in future system updates.
- Maintain consistent data entry practices across all branches to ensure that centralized reports and analytics remain accurate and useful.

5.5 Impact of the Study

The Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo

provides a practical digital solution to the operational challenges experienced by the business across its eight branches in Iloilo City. By combining digital sales processing,

centralized sales management, inventory management, customer management, promotions, and analytical reporting, the system helps organize important business information in one centralized environment.

For **Luna's Arrozcaldo**, the system reduces dependence on manual sales recording and physical collection of branch information. Sales transactions recorded through the digital POS can be stored centrally and connected with inventory updates, customer purchase history, promotions, and management reports. This gives management a more organized basis for monitoring branch performance and making informed business decisions.

For **branch managers and cashiers**, the system provides a more consistent and structured way of handling sales and inventory activities. It reduces handwritten calculations and recording while making important information easier to access. The centralized system also supports continuity of business information by providing a formal digital process instead of relying mainly on manual staff-to-staff information sharing.

The study also demonstrates how **digital technology can support a long-established heritage food business without changing its established identity and service practices**. Instead, technology is applied mainly to sales, inventory, customer, promotional, and management processes, strengthening the information foundation behind Luna's Arrozcaldo while supporting its continued operations.

5.6 Summary

The **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo** represents a direct response to the operational problems

identified in the study. By connecting a digital POS, centralized database, inventory management, customer management, promotions, and analytical reporting, the project establishes a unified information environment for an eight-branch food business.

The system's value lies not only in replacing paper with software but in connecting individual business activities into a common flow of information. A sale recorded at the POS can become part of the transaction history, inventory update, customer purchase history, promotion record, and management report. This connection provides the foundation for more organized operations and informed management.

The project also demonstrates the importance of developing technology around actual user requirements. Agile iterations, client review, role-based access, a normalized database, security controls, and user acceptance evaluation all support a system built around practical needs. Future improvements should continue to respond to verified operational requirements and user feedback.

In conclusion, the developed system provides Luna's Arrozcaldo with a structured digital foundation for modern sales processing and centralized management while remaining aligned with its established business operations in Iloilo City.